

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-III

TEST # 02

DATE : 05-08-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 02	Topics	DATE : 05 - 08 - 2018
PRE-MEDICAL NURTURE COURSE : PHASE - III		
PHYSICS	: Vectors, Unit & Dimension & Kinematics	
CHEMISTRY	: Atomic Structure	
BIOLOGY	: Cell : The Unit of Life, Cell Cycle and Cell Division, Biomolecules	

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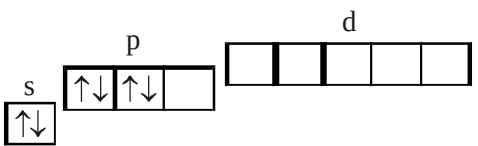
BEWARE OF NEGATIVE MARKING

1. If for two vectors $\vec{A}$ and $\vec{B}$, we have $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$:-
 (1) 0 (2) $\pi/4$ (3) $\pi/2$ (4) π
2. A pressure of 10^6 dynes/cm² is equivalent to
 (1) 10^5 N/m² (2) 10^4 N/m²
 (3) 10^6 N/m² (4) 10^7 N/m²
3. The velocity v of a particles is given in terms of time t by the equation.

$$v = at + \frac{b}{t+c}$$
 The dimension of a , b and c are
 (1) $L^2, T, L T^2$ (2) LT^2, LT, L
 (3) LT^{-2}, L, T (4) L, LT, T^2
4. kg m/s² stands for the unit of –
 (1) Energy (2) acceleration
 (3) Force (4) Momentum
5. The product (PV) has the dimensions –
 (1) $[ML^{-1} T^{-2}]$ (2) $[M^1 L^2 T^{-2}]$
 (3) $[M^1 L^2 T^{-2}]$ (4) $[M^1 L^2 T^{-3}]$
6. What is the physical quantity whose dimensions are $M L^2 T^{-2}$ –
 (1) Pressure (2) Work
 (3) Power (4) Momentum
7. Subtract 0.2 J from 5.27 J and express the result with correct number of significant figures -
 (1) 5.1 J (2) 5.06 J
 (3) 5.0 J (4) 5 J
8. Error in the measurement of radius of a sphere is 2%. Then error in the measurement of volume is -
 (1) 2% (2) 4% (3) 8% (4) 6%
9. In the relation $y = r \sin(\omega t - kx)$ the dimensions of $\frac{\omega}{k}$ are–
 (1) $[M^0 L^0 T^0]$ (2) $[M^0 L^1 T^{-1}]$
 (3) $[M^0 L^0 T^1]$ (4) $[M^0 L^1 T^0]$
10. The velocity v of waves produced in water depends on their wavelength λ , the density of water ρ , and acceleration due to gravity g . The square of velocity is proportional to -
 (1) $\lambda^{-1} g^{-1} \rho^{-1}$ (2) $\lambda^{\frac{1}{2}} g^{\frac{1}{2}}$
 (3) λg (4) $\lambda^2 g^{-2} \rho^{-1}$
11. The maximum error in the measurement of mass and length of the side of a cube are 2% and 1% respectively. The maximum error in its density is-
 (1) 2% (2) 1%
 (3) 3% (4) 5%
12. The equation $\frac{dv}{dt} = At - Bv$ is describing the rate of change of velocity of a body falling from rest in a resisting medium. The dimensions of A and B are -
 (1) LT^{-3}, T (2) LT^{-3}, T^{-1}
 (3) LT, T (4) LT, T^{-1}
13. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to n times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ is :-
 (1) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$ (2) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$
 (3) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$ (4) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$
14. If force F , acceleration A and time T are basic physical quantities, the dimensions of energy are –
 (1) $[F^2 A^{-1} T]$ (2) $[F A T^2]$
 (3) $[F A T^{-2}]$ (4) $[F A^{-1} T]$
15. If force (F), length (L) and time (T) be considered fundamental units, then units of mass will be –
 (1) $[F L^{-1} T^{-2}]$ (2) $[F^2 L T^{-2}]$
 (3) $[F L^{-1} T^2]$ (4) $[F L^{-2} T^{-1}]$

16. The time dependence of physical quantity P is given by $P = P_0 e^{-\alpha t^2 + \beta t + \gamma}$, where α, β, γ are constants and their dimensions are given by (where t is time) -
 (1) $M^0 L^0 T^{-2}, M^0 L^0 T^{-1}, M^0 L^0 T^0$
 (2) $M^0 L^{-1} T^{-2}, M^0 L^0 T^{-1}, M^0 L^0 T$
 (3) $M^0 L^0 T^{-1}, M L T^{-2}, M^0 L^0 T^{-1}$
 (4) $M L T, M L T^0, M^0 L^0 T^0$
17. Error in measurement of radius of a sphere is 1%. Then error in measurement of area is -
 (1) 2% (2) 3%
 (3) 4% (4) 5%
18. If $x = ab$, the maximum percentage error in the measurement of x will be -
 (1) $\left(\frac{\Delta a}{a} \times 100\%\right) \times \left(\frac{\Delta b}{b} \times 100\%\right)$
 (2) $\left(\frac{\Delta a}{a} \times 100\%\right) \div \left(\frac{\Delta b}{b} \times 100\%\right)$
 (3) $\left(\frac{\Delta a}{a} - \frac{\Delta b}{b}\right) \times 100\%$
 (4) $\left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right) \times 100\%$
19. The percentage errors in measurement of mass and speed are 4% and 3% respectively. The error in kinetic energy will be $[KE = \frac{1}{2}mv^2]$ -
 (1) 6% (2) 7% (3) 10% (4) 12%
20. The number of significant figures in 0.900 is -
 (1) 1 (2) 2 (3) 3 (4) 4
21. The velocity of body moving in one-dimension is given by $v = 2s + 3$, where s is displacement. Its acceleration is
 (1) 2v (2) 4v
 (3) v (4) $2(2s+3)$
22. The number of significant figures in (i) 0.03800 and (ii) 90.00 is -
 (1) (i) 4 (ii) 4 (2) (i) 2 (ii) 1
 (3) (i) 3 (ii) 3 (4) (i) 2 (ii) 4
23. The number of significant figures in 3.04×10^{23} is -
 (1) 2 (2) 3 (3) 23 (4) 25
24. The number of significant figures in $0.0060 \times 10^{-18} \text{ m}$ is -
 (1) 1 (2) 2 (3) 4 (4) 5
25. A particle moves along x-axis and its displacement at any time is given by $x(t) = 2t^3 - 3t^2 + 4t$ in SI units. The velocity of the particle when its acceleration is zero, is :-
 (1) 2.5 m s^{-1} (2) 3.5 m s^{-1}
 (3) 4.5 m s^{-1} (4) 8.5 m s^{-1}
26. A car moves from A to B with a speed of 30 kmph and from B to A with a speed of 20 kmph. What is the average speed of the car?
 (1) 50 kmph (2) 25 kmph
 (3) 10 kmph (4) 24 kmph
27. A particle starting with certain initial velocity and uniform acceleration covers a distance of 12 m in first 3 seconds and a distance of 30 m in next 3 seconds. The initial velocity of the particle is :-
 (1) 3 m s^{-1} (2) 2.5 m s^{-1}
 (3) 2 m s^{-1} (4) 1 m s^{-1}
28. If $\vec{A} = 2\hat{i} + \hat{j} + \sqrt{2}\hat{k}$, then find out the angle made by the vector from y-axis :-
 (1) $\cos^{-1}\left(\frac{2}{\sqrt{7}}\right)$ (2) $\cos^{-1}\left(\frac{1}{\sqrt{7}}\right)$
 (3) $\cos^{-1}\left(\frac{\sqrt{2}}{\sqrt{7}}\right)$ (4) $\cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$
29. The numerical value of the ratio of velocity to speed is :-
 (1) always less than one
 (2) always equal to one
 (3) always more than one
 (4) equal to or less speed
30. If the distance covered is zero, the displacement :-
 (1) must be zero
 (2) may or may not be zero
 (3) cannot be zero
 (4) depends upon the particle
31. An athlete completes one round of a circular track of radius R in 40 second. What will be his displacement at the end of 2 minute 20 second?
 (1) Zero (2) 2R
 (3) $2\pi R$ (4) $7\pi R$

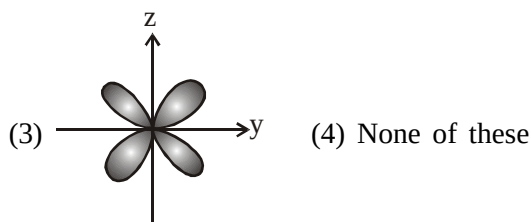
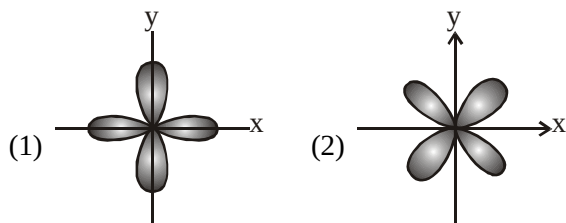
32. A body covers one-third of the distance with a velocity v_1 , the second one-third of the distance with a velocity v_2 and the remaining distance with a velocity v_3 . The average velocity is :-
- (1) $\frac{v_1 + v_2 + v_3}{3}$
 (2) $\frac{3v_1v_2v_3}{v_1v_2 + v_2v_3 + v_3v_1}$
 (3) $\frac{v_1v_2 + v_2v_3 + v_3v_1}{3}$
 (4) $\frac{v_1v_2v_3}{3}$
33. A particle experience constant acceleration for 20 seconds after starting from rest. If it travels a distance s_1 in the first 10 seconds and distance s_2 in the next 10 seconds, then :-
- (1) $s_2 = s_1$ (2) $s_2 = 2s_1$
 (3) $s_2 = 3s_1$ (4) $s_2 = 4s_1$
34. If x denotes displacement in time t and $x = a \cos t$, then acceleration is :-
- (1) $a \cos t$ (2) $-a \cos t$
 (3) $a \sin t$ (4) $-a \sin t$
35. A particle's position as a function of time is described as $y(t) = 2t^2 + 3t + 4$. What is the average velocity of the particle from $t = 0$ to $t = 3$ sec?
- (1) 3 m/s (2) 6 m/s
 (3) 9 m/s (4) 12 m/s
36. A particle moves along a straight line such that its displacement at any time t is given by :
 $s = (t^3 - 3t^2 + 2)t$
 The displacement when the acceleration becomes zero is :-
- (1) 0 m (2) 2 m (3) 3 m (4) -2 m
37. The velocity-time relation of an electron starting from rest is given by $u = kt$, where $k = 2 \text{ m/s}^2$. The distance traversed in 3 sec is :-
- (1) 9 m (2) 16 m (3) 27 m (4) 36 m
38. A vector $\vec{A}$ points vertically downward and $\vec{B}$ points toward south. The vector product $\vec{B} \times \vec{A}$ is :-
- (1) zero vector (2) along east
 (3) along west (4) along north
39. When 96.54 is divided by 2.40, the correct result is -
- (1) 40.2250 (2) 40.225
 (3) 40.23 (4) 40.2
40. When $\vec{A} \cdot \vec{B} = -AB$, then :-
- (1) $\vec{A}$ and $\vec{B}$ are perpendicular to each other
 (2) $\vec{A}$ and $\vec{B}$ act in the same direction
 (3) $\vec{A}$ and $\vec{B}$ act in the opposite direction
 (4) $\vec{A}$ and $\vec{B}$ can act in any direction
41. If $\vec{A} = (\hat{i} + \hat{j})$ and $\vec{B} = (\hat{j} + \hat{k})$ then value of $(\vec{A} \times \vec{B})$ is
- (1) $(\hat{i} + \hat{j} - \hat{k})$ (2) $(\hat{i} - \hat{j} + \hat{k})$
 (3) $(-\hat{i} + \hat{j} + \hat{k})$ (4) $(\hat{i} + \hat{j} + \hat{k})$
42. If $(\vec{A} \times \vec{B}) = (AB \sin \theta) \hat{n}$ then which of the following is incorrect
- (1) $\hat{n} \cdot \vec{A} = 0$ (2) $\hat{n} \cdot \vec{B} = 0$
 (3) $\hat{n} \cdot (\vec{A} \times \vec{B}) = 0$ (4) $\hat{n} \cdot (\vec{A} + \vec{B}) = 0$
43. The velocity of a particle is $\vec{v} = \hat{i} + \hat{j}$. The component of the velocity parallel to vector $(\hat{j} - \hat{k})$ is :-
- (1) $\frac{1}{2}(\hat{i} + \hat{j})$ (2) $\frac{1}{2}(\hat{j} + \hat{k})$
 (3) $\frac{-1}{2}(\hat{k} - \hat{j})$ (4) $\frac{1}{2}(\hat{i} + \hat{k})$
44. If $R = x\hat{i} + y\hat{j} + z\hat{k}$ makes α , β and γ angles with X-axis, Y-axis and Z-axis, respectively. Find the value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$.
- (1) zero (2) 1 (3) 2 (4) $\frac{1}{2}$
45. The angle between two vectors $\vec{A}$ and $\vec{B}$ is θ . Then the magnitude of the product $\vec{A} \cdot (\vec{B} \times \vec{A})$ is:-
- (1) $A^2 B$ (2) $A^2 B \sin \theta$
 (3) $A^2 B \sin \theta \cos \theta$ (4) Zero

46. Which of the following quantum number determines the orientation of the atomic orbital -
(1) n (2) ℓ (3) m (4) s
47. The maximum value of specific charge (e/m) of canal rays is for which gas -
(1) Helium (2) N_2 (3) CO_2 (4) H_2
48. Which of the following statement is not true for anode ray ?
(1) They are deflected by electric and magnetic fields
(2) Their e/m ratio depends on the gas in discharge tube used to produce anode rays
(3) The e/m ratio of anode rays is constant.
(4) They are produced by ionization of the gas in discharge tube
49. If kinetic energy of a proton is increased nine times the wavelength of the de-Broglie wave associated with it would become
(1) 3 times (2) 9 times
(3) $\frac{1}{3}$ times (4) $\frac{1}{9}$ times.
50. The number of electrons having $l = 0$ in chlorine atom ($Z = 17$) is
(1) 2 (2) 4 (3) 6 (4) 5.
51. As we go away from the nucleus, the difference between the successive energy levels -
(1) increases
(2) decreases
(3) remains constant
(4) first increases, then decreases
52. The ionization energy of hydrogen atom is 13.6 eV. What will be the ionization energy of He^+ ?
(1) 13.6 eV (2) 54.4 eV
(3) 122.4 eV (4) Zero
53. Calculate the wavelength of the spectral line when the electron in the hydrogen atom undergoes a transition from fourth energy level to second energy level ?
(1) 4.86 nm (2) 486.4 nm
(3) 48.6 nm (4) 4860 nm
54. The frequency of first line of Balmer series in hydrogen atom is ν_0 . The frequency of corresponding line emitted by singly ionised helium atom is
(1) $2\nu_0$ (2) $4\nu_0$ (3) $\nu_0/2$ (4) $\nu_0/4$
55. The ratio of ionization energy of H and Be^{+3} is
(1) 1 : 1 (2) 1 : 3 (3) 1 : 9 (4) 1:16
56. 
- The above configuration is not correct as it violates :
(1) Only Hund's rule
(2) Only Pauli's exclusion principle
(3) $(n + l)$ rule
(4) (Hund + Pauli) rule
57. **S-1:** The energy level of 3d subshell is lesser than 4s for multielectronic system.
S-2 : The energy level of 3d subshell is higher than 4s for unelectronic system.
(1) S-1 and S-2 both are true
(2) S-1 is true and S-2 is false
(3) S-1 and S-2 both are false
(4) S-1 is false and S-2 is true
58. What are the values of the orbital angular momentum of an electron in the orbitals 1s, 3s, 3d and 2p?
(1) 0, 0, $\frac{h}{2\pi}\sqrt{6}$, $\frac{h}{2\pi}\sqrt{2}$
(2) 1, 1, $\frac{h}{2\pi}\sqrt{4}$, $\frac{h}{2\pi}\sqrt{2}$
(3) 0, 1, $\frac{h}{2\pi}\sqrt{6}$, $\frac{h}{2\pi}\sqrt{3}$
(4) 0, 0, $\frac{h}{2\pi}\sqrt{20}$, $\frac{h}{2\pi}\sqrt{6}$
59. Bohr's model can explain-
(1) the spectrum of hydrogen atom only
(2) spectrum of an atom or ion containing one electron only
(3) the spectrum of hydrogen molecule
(4) the solar spectrum

60. The correct ground state electronic configuration of chromium atom is-

- (1) [Ar] $3d^5 4s^1$ (2) [Ar] $3d^4 4s^2$
(3) [Ar] $3d^6 4s^0$ (4) [Ar] $4d^5 4s^1$

61. Shape of d_{xy} orbital is :-



62. The number of radial nodal surface in 3s and 2p

- (1) 2, 0 (2) 2, 1
(3) 1, 0 (4) 0, 2

63. If uncertainty in position and momentum are equal, then uncertainty in velocity is -

- (1) $\sqrt{\frac{h}{\pi}}$ (2) $\sqrt{\frac{h}{2\pi}}$
(3) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (4) None

64. The nuclides $^{40}_{18}\text{Ar}$ and $^{41}_{19}\text{K}$ are -

- (1) isotopes (2) isobars
(3) isotones (4) isoelectronic

65. How many moles are present in 9.8 mg of H_2SO_4 (molecular Mass of $\text{H}_2\text{SO}_4 = 98 \text{ amu}$) -

- (1) 10^{-3} (2) 10^{-4}
(3) 10^{-2} (4) 10^{-5}

66. Which of the following orbital does not make sense?

- (1) 3d (2) 2f (3) 5p (4) 7s

67. The number of spherical nodes in 4s orbital is :-

- (1) 4 (2) ∞ (3) 2 (4) 3

68. If the threshold wavelength (λ_0) for ejection of electron from metal is 330 nm, then work function for the photoelectric emission is :-

- (1) $1.2 \times 10^{-18} \text{ J}$ (2) $1.2 \times 10^{-20} \text{ J}$
(3) $6 \times 10^{-19} \text{ J}$ (4) $6 \times 10^{-12} \text{ J}$

69. How many unpaired electrons are there in Ni^{2+} ? (Atomic no. of Ni = 28)

- (1) 0 (2) 2 (3) 4 (4) 8

70. Any p orbital can accommodate up to

- (1) Four electrons
(2) Two electrons with parallel spin
(3) Six electrons
(4) Two electrons with opposite spin

71. The radius of the second Bohr orbit for Li^{2+} is :-

- (1) $0.529 \times \frac{4}{3} \text{ \AA}$ (2) $0.529 \times \frac{2}{3} \text{ \AA}$
(3) $0.529 \times \frac{4}{9} \text{ \AA}$ (4) $0.529 \times \frac{2}{9} \text{ \AA}$

72. The ratio of kinetic energy and total energy of an electron in a Bohr orbit of a hydrogen-like species is :-

- (1) $1/2$ (2) $-1/2$ (3) 1 (4) -1

73. If velocity of an electron in 1st orbit of H atom is V, what will be the velocity in 3rd orbit of Li^{2+} ?

- (1) V (2) $\frac{V}{3}$ (3) 3V (4) 9V

74. The spectral line obtained when an electron jumps from $n = 6$ to $n = 2$ level in hydrogen atom belongs to the :-

- (1) Balmer series (2) Lyamn series
(3) Paschen series (4) Pfund series

75. The sum of the number of neutrons and proton in all the isotope of hydrogen is :-

- (1) 6 (2) 5 (3) 4 (4) 3

76. Number of angular nodes in 3d orbital is :-

- (1) 3 (2) 2 (3) 1 (4) zero

77. In which region of electromagnetic radiations does balmer series fall?

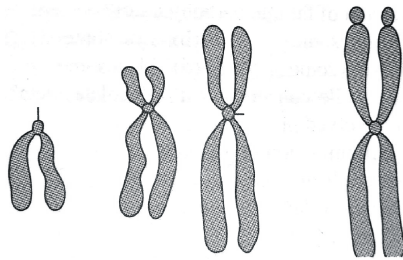
- (1) UV region (2) IR region
(3) Visible region (4) Far IR region

78. Which orbitals are spherically symmetrical ?
 (1) d-orbitals (2) p-orbital
 (3) f-orbital (4) s-orbital
79. Within a given shell, correct order of energies for the s, p, d and f subshell is :-
 (1) $f > d > s > p$
 (2) $s > p > f > d$
 (3) $f > d > p > s$
 (4) $s > f > p > d$
80. Wavelength of last line of lyman series of 'H' atom is :-
 (1) 912 Å (2) 1824 Å
 (3) 456 Å (4) 109700 Å
81. Value of n_1 for pfund series is :-
 (1) $n_1 = 4$ (2) $n_1 = 2$
 (3) $n_1 = 3$ (4) $n_1 = 5$
82. The wavenumber of a spectral line for a given transition is $X\text{cm}^{-1}$ for He^+ ion. Then its value for Be^{+3} ion (isoelectronic of He^+) for same transition is :-
 (1) $X\text{cm}^{-1}$ (2) $4X\text{cm}^{-1}$
 (3) $\frac{X}{4}\text{cm}^{-1}$ (4) $2X\text{cm}^{-1}$
83. Which of the following sets of quantum numbers can be correct for an electron in 4f-orbital :-
 (1) $n = 4, l = 3, m = -2, s = 0$
 (2) $n = 4, l = 3, m = +4, s = -\frac{1}{2}$
 (3) $n = 4, l = 3, m = 1, s = +\frac{1}{2}$
 (4) $n = 4, l = 2, m = +1, s = +\frac{1}{2}$
84. Which 'd' orbital does not have four lobes :-
 (1) $d_{x^2-y^2}$ (2) d_{xy}
 (3) d_{yz} (4) d_{z^2}
85. The modern periodic table has :-
 (1) Eight horizontal rows and seven vertical columns
 (2) Seven horizontal rows and eighteen vertical columns
 (3) Seven horizontal rows and seven vertical columns
 (4) Eight horizontal rows and eight vertical columns
86. de-Broglie equation is :-
 (1) $n\lambda = 2\sin\theta$ (2) $E = h\nu$
 (3) $\Delta x \cdot \Delta p \geq \frac{h}{4\lambda}$ (4) $\lambda = \frac{h}{mv}$
87. The fourth line in Balmer series corresponds to an electronic transition between which Bohr's orbit in hydrogen
 (1) $5 \rightarrow 3$ (2) $6 \rightarrow 2$
 (3) $6 \rightarrow 3$ (4) $5 \rightarrow 2$
88. When atoms are bombarded with alpha particles, only a few in million suffer deflection, others pass out undeflected. This is because :-
 (1) The force of repulsion on the moving alpha particle is small
 (2) The force of attraction on the alpha particle to the oppositely charged electrons is very small.
 (3) There is only one nucleus and large number of electrons
 (4) The nucleus occupies much smaller volume compared to the volume of the atom.
89. Which of the following have the maximum frequency :-
 (1) Cosmic rays (2) X-rays
 (3) Radiowaves (4) Infra red rays
90. The difference between the wavenumber of 1st line of Balmer series and last line of paschen series for Li^{+2} ion is :-
 (1) $\frac{R}{36}$ (2) $\frac{5R}{36}$
 (3) $4R$ (4) $\frac{R}{4}$

- 91.** Select the incorrect match :-
 (1) Mesosomes – Cellular respiration
 (2) Pili – Nutrition
 (3) Fimbriae – Attachment
 (4) Plasmid – Extrachromosomal genetic material
- 92.** The organelle common in both prokaryotic and eukaryotic cell is :-
 (1) Plastid (2) Ribosomes
 (3) Plasmid (4) DS linear DNA
- 93.** Ribosomes in prokaryotes
 (A) Are 70S type
 (B) Not bounded by any membrane
 (C) They have 50 S and 30 S subunit
 (D) Form polysomes
 (1) (A), (B), (C) are correct
 (2) Only (C) is correct
 (3) All are correct
 (4) (A), (B) and (D) are correct
- 94.** The term glycocalyx is used for :-
 (1) A layer surrounding the cell wall of bacteria
 (2) A layer present between cell wall and cell membrane of bacteria
 (3) Cell wall of bacteria
 (4) A layer present inner side to cell membrane
- 95.** Prokaryotic cells are different from Eukaryotic cells because having :-
 (1) 80S ribosome
 (2) DS linear DNA with histone protein
 (3) DNA with high GC content
 (4) Well organised nucleus
- 96.** A protoplast is :-
 (1) A cell organelle
 (2) A cell without a cell wall
 (3) A cell without nucleus
 (4) A cell with a cell wall
- 97.** Cell wall is absent in :-
 (1) Animal cell (2) Amoeba
 (3) Mycoplasma (4) All of these
- 98.** Which of the following is present in plant cell but absent in animal cell ?
 (1) Cell wall (2) Large sap vacuole
 (3) Plastids (4) All of these
- 99.** Cell wall :-
 (1) Rigid structure of the cell
 (2) Permeable
 (3) Provide barrier to undesirable and large macromolecules
 (4) All of these
- 100.** Select the mismatched pair.
 (1) Sap vacuole – excretory product
 (2) Gas vacuole – buoyancy / Help in floating
 (3) Contractile vacuole – photosynthesis
 (4) Food vacuoles – phagocytosis
- 101.** Blue and red colour of petals is mainly due to :-
 (1) Anthocyanin
 (2) Fucoxanthin
 (3) Phycoerythrin
 (4) Haemoglobin
- 102.** Cellulose, hemicellulose, pectins and proteins are possible in the cell wall of :-
 (1) Bacteria (2) Higher plants
 (3) Green algae (4) Fungi
- 103.** Which of the following plastid store protein?
 (1) Aleuroplast (2) Amyloplast
 (3) Chromoplast (4) Elaioplast
- 104.** Mitochondria and chloroplasts are :-
 (1) Considered as part of the endomembrane system
 (2) Single membrane bound cell organelles
 (3) Divide by binary fission
 (4) Having 80 S ribosomes
- 105.** Ribosomes of mitochondria, chloroplast and R.E.R are respectively :-
 (1) 80S, 70S, 70S
 (2) 80S, 80S, 80S
 (3) 70S, 70S, 80S
 (4) 70S, 80S, 70S
- 106.** Which of the following has direct connections to outer membrane of nuclear membrane?
 (1) RER (2) Golgi bodies
 (3) Lysosomes (4) Mitochondria
- 107.** Main function of dictyosome is :-
 (1) Respiration (2) Storage
 (3) Secretion (4) Breakdown of fats

- 108.** The cementing layer between two adjacent plants cells is :-
 (1) Primary wall (2) Middle lamella
 (3) Secondary wall (4) Plasma membrane
- 109.** Plasma membrane is regarded as asymmetric because :-
 (1) Carbohydrate in outer and spectrin protein inner sides are present
 (2) It contains carbohydrate
 (3) Protein icebergs in sea of lipid
 (4) Made up of glycoprotein and lipid
- 110.** Choose the incorrect match
 (1) Cristae – Infolding of inner membrane of mitochondria
 (2) Granna – Stacks of thylakoids
 (3) Ribosome – Protein synthesis
 (4) Elaioplasts – Store protein granules
- 111.** In human being, the RBC membrane has approximately :-
 (1) 52% protein, 40% lipid
 (2) 40% protein, 52% lipid
 (3) 42% protein, 50% lipid
 (4) 30% protein, 50% lipid
- 112.** Choose correct one w.r.t. composition of algae cell wall :-
 (1) Cellulose + Galactans + Mannans + Minerals like CaCO_3
 (2) Hemicellulose and pectin
 (3) Cellulose + Hemicellulose + Pectin + Minerals like CaCO_2
 (4) Suberin + Cellulose + Galactans + Silica
- 113.** Cathepsin enzyme is helpful in digestion of tail of tadpole larva, it is secreted by :-
 (1) Sphaerosome (2) Peroxysome
 (3) Lysosome (4) Glyoxysome
- 114.** Which of the following is not a part of lysosome?
 (1) Primary lysosome
 (2) Diplosome
 (3) Heterophagosome
 (4) Autophagic vacuole
- 115.** The unit for measurement of sedimentation of ribosome is :-
 (1) Angstrom ($\text{\AA}$)
 (2) Micron (μ)
 (3) Svedberg unit (S)
 (4) Nano meter (ηm)
- 116.** The principle protein of Cilia and Flagella is :-
 (1) Tubulin
 (2) Albumin
 (3) Globulin
 (4) Fibrinogen
- 117.** Function of centriole is :-
 (1) Formation of spindle fibre
 (2) Formation of Nucleolus
 (3) Initiation of cell membrane
 (4) Formation of cell plate
- 118.** The site of synthesis and degradation of H_2O_2 in a plant cell is :-
 (1) Lysosome
 (2) Sphaerosome
 (3) Peroxisome
 (4) Microsome
- 119.** A living plant cell which becomes enucleated at maturity is :-
 (1) Sieve tube
 (2) Vessel
 (3) Tracheid
 (4) Collenchyma cell
- 120.** Micro tubules are present in all, except :-
 (1) Flagella
 (2) Spindle fibres
 (3) Centrioles
 (4) Microbodies
- 121.** An elaborate network of filamentous and proteinaceous structures present in the cytoplasm of an eukaryotic cell is called :-
 (1) Cytoskeleton
 (2) Chromatin fibres
 (3) Polysome
 (4) Microsome

122. Mark the correct sequence of the type of chromosomes given below :-



- (1) Acrocentric, Acentric, Submetacentric, Telocentric
 - (2) Telocentric, Acrocentric, Metacentric, Submetacentric
 - (3) Telocentric, Acrocentric, Submetacentric, Metacentric
 - (4) Acentric, Metacentric, Acrocentric, Telocentric
123. Disc shaped structure present on the side of primary constriction meant for spindle fibre attachment are called :-
- (1) Secondary constriction
 - (2) Centriole
 - (3) Kinetochore
 - (4) Diplosome
124. Nucleus was first observed by Robert Brown in :-
- (1) Maize root
 - (2) *Acetabularia*
 - (3) Orchid root
 - (4) *Fritillaria*
125. Both cilium and flagellum of a Eukaryotic cell emerge from a structure which shows :-
- (1) 9 + 2 microtubule organisation
 - (2) Cart wheel like organisation of microtubule
 - (3) A pair of centrally located microtubule
 - (4) Unit membrane bound structure
126. Nucleolus is the main site for the synthesis of :-
- (1) Ribosomal subunits
 - (2) Proteins
 - (3) Euchromatin
 - (4) Heterochromatin
127. Centrosome is an organelle containing two cylindrical structure :-
- (1) Cisternae
 - (2) Tubules
 - (3) Centrioles
 - (4) Centromere

128. Hub is connected with microtubules of the peripheral triplet in :-
- (1) Basal body of bacterial flagellum
 - (2) Centriole
 - (3) Axoneme of both Eukaryotic and Prokaryotic flagella
 - (4) All of these
129. Which of the following organelles is involved in the conversion of fat into carbohydrates?
- (1) Peroxisome
 - (2) Lysosome
 - (3) Glyoxysome
 - (4) Mesosome
130. Non membranous organelle without DNA with duplicating nature is :-
- (1) Plasmid
 - (2) Centriole
 - (3) Ribosome
 - (4) Plastid
131. The sequence of cell cycle is :-
- (1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
 - (2) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$
 - (3) $G_1 \rightarrow G_2 \rightarrow M \rightarrow S$
 - (4) $G_1 \rightarrow M \rightarrow G_2 \rightarrow S$
132. In stage of cell cycle, the cell remain metabolically active but do not divide called.
- (1) G_0
 - (2) G_2
 - (3) S
 - (4) G_1
133. A human cell divide through cell cycle in :-
- (1) 9 hours
 - (2) 90 minutes
 - (3) 24 hours
 - (4) 90 hours
134. Choose the incorrect pair.
- (1) G_1 – Post mitotic gap phase
 - (2) G_0 – Quiescent phase
 - (3) G_2 – Histone protein synthesis
 - (4) S – DNA synthesis
135. At which stage of cell cycle are histone protein synthesised in a eukaryotic cell?
- (1) During G_2 phase of interphase
 - (2) During S phase
 - (3) During entire interphase
 - (4) During telophase of M phase

- 136.** Content of nucleic acid in protoplasm is :-
 (1) 70% (2) 15% (3) 5% (4) 3%
- 137.** All of the following are found in acid insoluble fraction, except :-
 (1) Proteins
 (2) Nucleic acid
 (3) Polysaccharide
 (4) Amino acid
- 138.** If we observe the average composition of cell, then :-
 (1) Water is the most abundant constituent
 (2) Percentage of carbohydrate is more than lipid
 (3) Protein constitute 10–15%
 (4) All are correct
- 139.** Biomolecules :-
 (1) Are small sized
 (2) Have molecular weight less than 1000 daltons
 (3) Are found in acid soluble fraction
 (4) All are correct
- 140.** Which of the following is not a secondary metabolite?
 (1) Rubber (2) Carotenoids
 (3) Monoterpenes (4) Sugars
- 141.** Which of the following is considered as macromolecule, though they have molecular weight less than 800 daltons?
 (1) Proteins (2) Lipids
 (3) Polysaccharide (4) Nucleic acid
- 142.** Which of the following is not a secondary metabolite?
 (1) Sucrose (2) Abrin
 (3) Morphine (4) Anthocyanin
- 143.** Which of the following set contains monosaccharide only?
 (1) Lactose and glucose
 (2) Fructose and ribose
 (3) Glucose and sucrose
 (4) Glucose and maltose
- 144.** Which of the following set consists only homo polysaccharides?
 (1) Glycogen and peptidoglycan
 (2) Starch and inulin
 (3) Glycogen and heparin
 (4) Starch and hyaluronic acid
- 145.** Which of the following polysaccharides contain α 1 – 4 and α 1 – 6 glycosidic linkage?
 (A) Starch
 (B) Glycogen
 (C) Chitin
 (D) Cellulose
 (1) A and C (2) A and B
 (3) B and C (4) C and D
- 146.** Which of the following polysaccharides is called as animal starch?
 (1) Glycogen (2) Inulin
 (3) Amylose (4) Chitin
- 147.** The most abundant organic compound of the plant kingdom :-
 (1) Chitin (2) Collagen
 (3) Cellulose (4) Starch
- 148.** Which one of the following is **correct** for raffinose w.r.t. their sugar components?
 (1) Galactose – glucose – fructose
 (2) Galactose – glucose – mannose
 (3) Galactose – trehalose – fructose
 (4) mannose – glucose – fructose
- 149.** Type of linkage present amylopectin is :-
 (1) β 1–4 (2) α 1–4, α 1–6
 (3) β 1–6 (4) α 1–4
- 150.** Hydrolysis of maltose yields :-
 (1) Glucose + glucose
 (2) Fructose only
 (3) Galactose + glucose + fructose
 (4) Galactose + galactose + glucose + fructose

151. The sweetest naturally occurring sugar is :-

- (1) Sucrose (2) Fructose
(3) Maltose (4) Lactose

152. Milk sugar is :-

- (1) Lactose (2) Sucrose
(3) Maltose (4) Fructose

153. Glucose is :-

- (1) blood sugar (2) grape sugar
(3) Dextrose (4) all of these

154. Which of the following is not a function of polysaccharides?

- (1) Energy storage
(2) Structural support
(3) Energy source
(4) Storage of genetic information

155. Zwitter ion has :-

- (1) Only positive charge
(2) Only negative charge
(3) Either positive or negative charge
(4) Both positive and negative charge

156. Which is the most abundant protein in animal world and in the whole biosphere?

- (1) Insulin and Collagen
(2) Albumin and globin
(3) Collagen and RuBisCo
(4) Trypsin and GLUT-4

157. The simplest amino acid is :-

- (1) Lysine (2) Glycine
(3) Valine (4) Glutamic acid

158. Which of the following is a sulphur containing amino acid?

- (1) Serine (2) Proline
(3) Cysteine (4) Alanine

159. Which of the following bonds is used for stabilising the primary structure of protein?

- (1) Hydrogen bond
(2) Peptide bond
(3) Hydrophobic interaction
(4) Disulphide bond

160. _____ structure gives the protein a three dimensional conformation.

- (1) Primary (2) Secondary
(3) Tertiary (4) None of these

161. A β -pleated sheet organization in a polypeptide is an example of :-

- (1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) Quaternary structure

162. Which of the following statements is not true?

- (1) Galactose is a disaccharide present in milk
(2) Lecithin is a phospholipid present in cell membrane
(3) Lipids are not strictly macromolecules
(4) PUFA are least harmful for heart

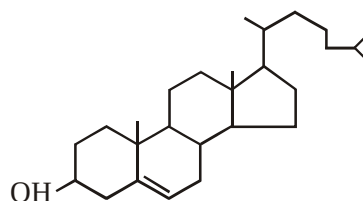
163. Which of the following is a saturated fatty acid with 16 carbon atoms?

- (1) Palmitic acid
(2) Stearic acid
(3) Arachidic acid
(4) Oleic acid

164. Lipids are :-

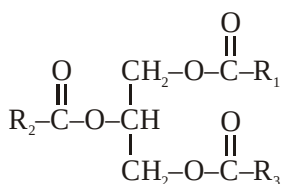
- (1) Fats or related substances
(2) Insoluble in water
(3) Soluble in organic solvents like acetone, benzene, etc.
(4) All are correct

165. The given structure is of



- (1) Lecithin (2) Palmitic acid
(3) Cholesterol (4) Ribose

166. This is the structure of



- (1) Steroid (2) Triglyceride
(3) Cholesterol (4) Lecithin

167. The biomolecules which provide stability to biological membrane as well as it is precursor of Vitamin-D in animals is :-

- (1) Lecithin (2) Cholesterol
(3) Wax (4) Stearic acid

168. Which element is common in haemoglobin and myoglobin ?

- (1) Fe (2) Cu
(3) Mn (4) Mg

169. Glycogen is stored in :-

- (1) Liver and muscles
(2) Liver only
(3) Muscles only
(4) Pancreas

170. Which one is a disaccharide?

- (1) Galactose (2) Fructose
(3) Maltose (4) Dextrin

171. To get quick energy one should use :-

- (1) Carbohydrate (2) Fats
(3) Vitamins (4) Proteins

172. Glycogen is :-

- (1) Polymer of amino acids
(2) Polymer of fatty acids
(3) Unsaturated fats
(4) Polymer of glucose

173. Carbohydrate is :-

- (1) Polymers of fatty acid
(2) Polymer of amino acids
(3) Polyhydroxy aldehyde or ketone
(4) None

174. The amino acids which are not synthesized in the body are called :-

- (1) Non-essential
(2) Essential
(3) Deaminated
(4) All of them

175. True statement for cellulose molecule :-

- (1) β - 1' - 4" linkage. unbranched
(2) β - 1' - 4" linkage. branched
(3) α - 1' - 4" linkage. branched
(4) β - 1' - 6" linkage. unbranched

176. Amylose and Amylopectin chains occur in :-

- (1) Glycogen (2) Starch
(3) Cellulose (4) Chitin

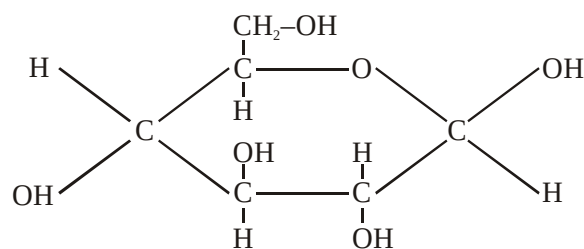
177. One of the major components of cell wall of most fungi is :-

- (1) Chitin
(2) Peptidoglycan
(3) Cellulose
(4) Hemicellulose

178. A typical fat molecule is made up of :-

- (1) Three glycerol molecules and one fatty acid molecule
(2) One glycerol and three fatty acid molecules
(3) One glycerol and one fatty acid molecule
(4) Three glycerol and three fatty acid molecules

179. Following structure is :-



- (1) α -Glucose (2) β -Glucose
(3) Ribose (4) Deoxyribose

180. Which of the following bond is/are found in tertiary structure of protein :-

- (1) peptide bond & hydrogen bond
(2) disulphide & hydrophobic bond
(3) ionic bond
(4) all the above

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